Capstone Project 1 Report

"Automated Attendance Tracking System using Power BI "

Domain: Business Intelligence & Automation Systems

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1. Introduction

Attendance management plays a vital role in monitoring discipline, productivity, and compliance in both colleges and companies. Traditional attendance methods such as manual registers or spreadsheets are inefficient, error-prone, and time-consuming. With the availability of data visualization and business intelligence tools, organizations can now automate attendance tracking and gain real-time insights. This project focuses on developing an Automated Attendance Tracking System using Power BI to visualize attendance data, monitor trends, and support decision-making.

2. Problem Statement

Manual and semi-automated attendance systems suffer from several limitations:

- Inaccurate and inconsistent data
- Difficulty in tracking attendance trends over time
- Time-consuming report generation
- No centralized, visual monitoring system
- Limited insights for management and administrators

There is a need for a Power BI-based automated system that converts raw attendance data into interactive dashboards and meaningful insights.

3. Objective

The main objectives of this project are:

- To automate the attendance analysis using Power BI
- To visualize daily, monthly, and yearly attendance
- To identify absenteeism and late trends
- To reduce manual reporting efforts
- To support administrative and managerial decision-making

4. Project Domain

Domain : Information technology – Business Intelligence & Automation systems

Sub-domain : Attendance Management System / Data Analytics

5. Tools & Technologies

Tool	Purpose
Power BI	Dashboard creation and data visualization
MS Excel / CSV	Source attendance data
Power Query	Data cleaning and transformation
DAX	Measures and calculated fields

6. Methodology

1. **Data Collection :**

   Attendance data was collected in Excel / CSV format containing employee/student details, dates, and attendance status.

2. **Data Cleaning (Power Query) :**
   - Removed duplicates and null values
   - Standardized data formats
   - Created attendance status columns (present, absent, late)
3. **Data Modelling :**
   - Established relationships between date, department, and user tables
   - Optimized model for reporting
4. **DAX :**
   - Total Count
   - Present Count
   - Absent Count
   - Attendance Percentage
   - Late Attendance Count
5. **Dashboard Development :**

   Created interactive visuals using slicers and filters.

6. Data Description

Key Fields:

- Employee ID
- Name
- Date
- Attendance status (present / absent / late)
- Performance
- Salary

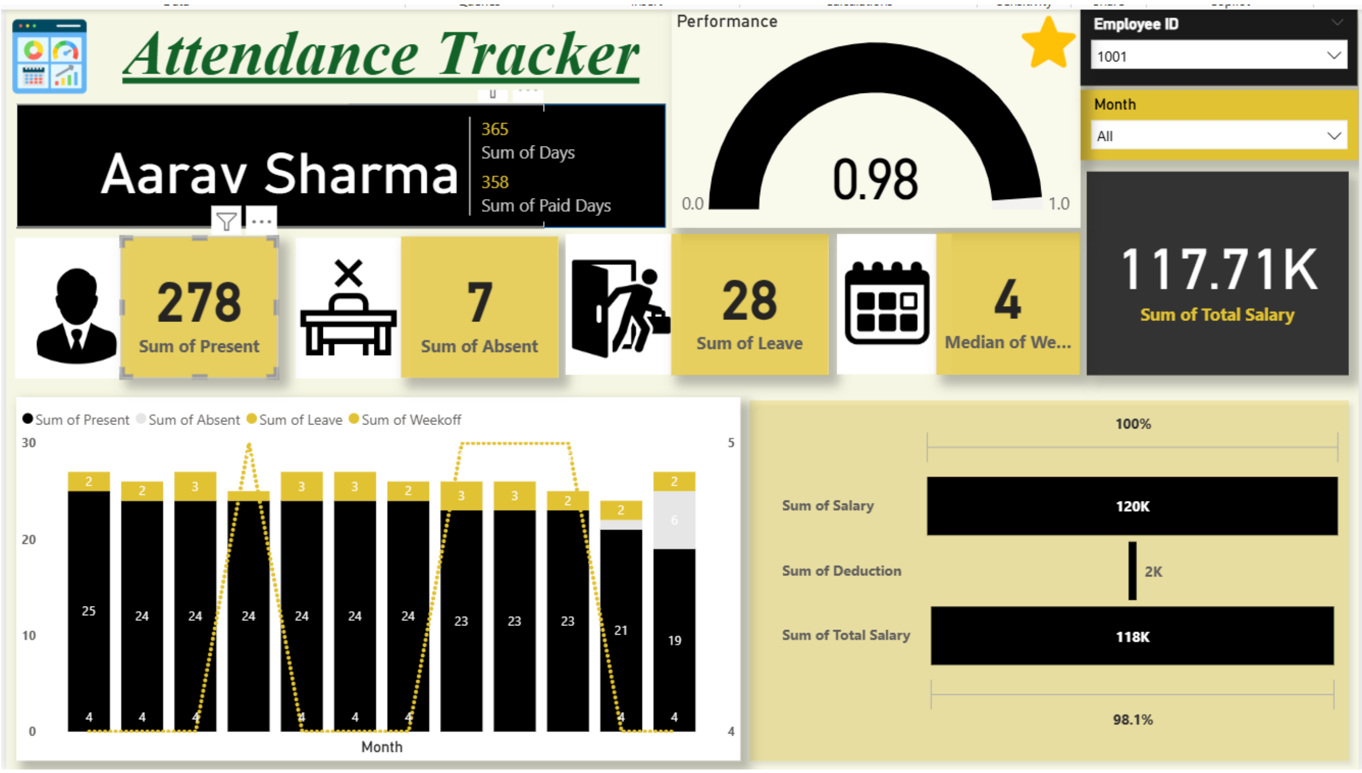
7. Dashboard Design

Sections:

1. The **Header** section displays the employee name, total working days, and total paid days.
2. A **Performance Guage** shows attendance efficiency (0.98), making it easy to evaluate employee performance.
3. **KPI Cards** presents key metrics such as total present days, absent days, leave days and median week-offs.
4. A **monthly Attendance Chart** shows present, absent, leave and week-offs trends across months.
5. The **Salary Analysis** section displays total salary, deductions, and net payable salary with percentage efficiency.
6. **Slicers** for Employee ID and month allow interactive and dynamic analysis.

This dashboard enables quick monitoring of attendance patterns, reduces manual efforts, and support HR decision-making.

Dashboard :



8. Key Insights

1. Attendance percentage fluctuates during specific periods
2. Late arrivals are more frequent on particular days
3. Overall attendance performance can be monitored in real time

9. Advantages

- Fully automated attendance analysis
- Real-time interactive dashboards
- Easy identification of attendance issues
- Reduced manual efforts
- Scalable for colleges and companies

10. Challenges Faced

- Handling inconsistent raw attendance data
- Creating accurate Dax measures
- Designing user-friendly dashboards
- Ensuring data accuracy during refresh

11. Real-Life Applications

This dashboard can be used in:

- Student attendance monitoring in colleges
- Employee attendance tracking in companies
- HR and academic performance analysis
- Compliance and audit reporting

12. Conclusion

The Automated Attendance Tracking System using Power BI successfully transformed raw attendance data into meaningful visual insights. The dashboard enables administrators and management to track attendance efficiently, identify patterns, and make informed decisions. This project demonstrates the effective use of Power BI in real-world automation and analytics scenarios.

13. Future Scope

- Integration with biometric or face recognition systems
- Real-time database connectivity
- Mobile-friendly dashboards
- Predictive analysis for absenteeism
- Cloud deployment with Power BI service

14. References

- Microsoft Power BI Documentation
- Data Analytics and BI Learning Resources